
DEPARTMENT OF MATHEMATICS

Mathematics Lab Assessment No. 1

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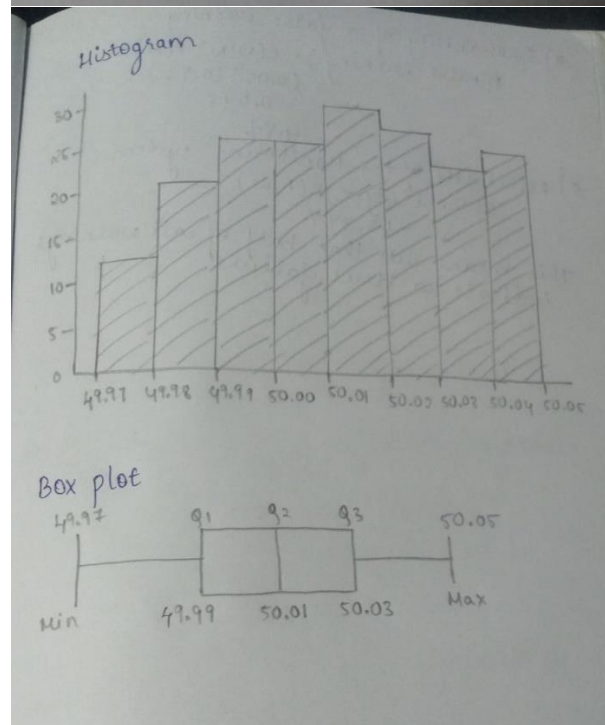
Question	A CNC milling machine is programmed to manufacture a precision shaft with a nominal diameter of: Target Diameter = 50.00 mm To evaluate machining consistency, the diameter of shafts produced during 200 consecutive production cycles is measured using a calibrated digital micrometer. Using the given dataset: • Calculate the mean diameter and mean error from the target value. • Determine the median, variance, standard deviation, and coefficient of variation. • Calculate the first quartile (Q1), second quartile (Q2), third quartile (Q3), and interquartile range (IQR). • Plot a histogram and box plot of the data. • Identify any outliers using the IQR method. Based on your statistical analysis, comment on: • The accuracy of the CNC machining process. • The precision/repeatability of the process. • Whether the process appears statistically stable. • Whether the machine is capable of maintaining the target diameter of 50.00 mm $\pm$ 0.05 mm.
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Solution i) Identify the parameters and mathematical concept

Parameter	Symbol
Diameter Measurement	x
Target Diameter	T
Number of Observations	n
Mean	$\bar{x}$
Median	Q2
Variance	(s ²)
Standard Deviation	s
Coefficient of Variation	CV
Quartiles	Q1, Q2, Q3
Interquartile Range	IQR

ii) Solve analytically

Mean $\bar{x} = 50.0087 \text{ mm}$
 Mean error $= \bar{x} - T$
 $= 0.0087 \text{ mm}$
 Median $= 50.01 \text{ mm}$
 Variance $s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$
 $= 0.000469 \text{ mm}^2$
 Standard deviation
 $s = 0.02166 \text{ mm}$
 Coefficient of variation
 $CV = \frac{s}{\bar{x}} \times 100$
 $= 0.0433\%$
 $Q_1 = 49.99 \text{ mm}$
 $Q_2 = 50.01 \text{ mm}$
 $Q_3 = 50.03 \text{ mm}$
 $IQR = Q_3 - Q_1$
 $= 0.04 \text{ mm}$
 Lower bound $= Q_1 - 1.5(IQR)$
 $= 49.93$
 Upper bound $= Q_3 + 1.5(IQR)$
 $= 50.09$
 All observations lie between 49.93 and 50.09 so NO outliers.



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iii) GeoGebra Screenshot / Program Execution

```
import numpy as np
import matplotlib.pyplot as plt

data = [
49.98, 50.02, 50.01, 49.99, 50.03, 49.97, 50.04, 50.00, 50.01, 49.98,
50.02, 49.99, 50.03, 50.00, 49.97, 50.05, 50.01, 49.99, 50.02, 49.98,
50.01, 50.03, 49.99, 50.00, 50.02, 49.98, 50.04, 50.01, 49.99, 50.03,
50.00, 50.02, 49.97, 50.01, 50.04, 49.99, 50.02, 50.00, 50.03, 49.98,
50.01, 49.99, 50.04, 50.02, 49.97, 50.03, 50.00, 50.01, 49.98, 50.02,
50.04, 49.99, 50.01, 50.03, 50.00, 49.98, 50.02, 50.01, 49.99, 50.05,
50.00, 50.03, 49.97, 50.02, 50.01, 49.98, 50.04, 50.00, 49.99, 50.03,
50.02, 49.98, 50.01, 50.04, 50.00, 49.99, 50.02, 50.03, 49.97, 50.01,
50.00, 50.02, 49.99, 50.04, 50.01, 49.98, 50.03, 50.00, 49.99, 50.02,
50.04, 50.01, 49.97, 50.03, 50.00, 50.02, 49.98, 50.01, 50.04, 49.99,
50.03, 50.00, 50.01, 49.98, 50.02, 50.04, 49.99, 50.03, 50.00, 50.01,
49.97, 50.02, 50.04, 49.99, 50.03, 50.01, 50.00, 49.98, 50.02, 50.05,
50.01, 49.99, 50.04, 50.02, 49.98, 50.03, 50.00, 50.01, 49.99, 50.04,
50.02, 49.97, 50.03, 50.01, 50.00, 49.98, 50.02, 50.04, 49.99, 50.01,
50.03, 50.00, 49.98, 50.02, 50.05, 49.99, 50.01, 50.04, 50.00, 50.03,
49.97, 50.02, 50.01, 49.99, 50.04, 50.00, 50.03, 49.98, 50.01, 50.02,
50.04, 49.99, 50.00, 50.03, 49.98, 50.01, 50.05, 50.02, 49.97, 50.04,
50.00, 50.01, 49.99, 50.03, 50.02, 49.98, 50.04, 50.00, 50.01, 49.99,
50.03, 50.02, 49.97, 50.04, 50.00, 50.01, 49.98, 50.03, 50.02, 49.99
]
```

```
target = 50.00

# Descriptive Statistics
mean = np.mean(data)
median = np.median(data)
variance = np.var(data, ddof=1)
std_dev = np.std(data, ddof=1)
cv = (std_dev / mean) * 100

# Quartiles
q1 = np.percentile(data, 25)
q2 = np.percentile(data, 50)
q3 = np.percentile(data, 75)
iqr = q3 - q1

# Mean Error
mean_error = mean - target

# Outlier Detection using IQR
lower_bound = q1 - 1.5 * iqr
upper_bound = q3 + 1.5 * iqr

outliers = [x for x in data if x < lower_bound or x > upper_bound]

# Display Results
print("Mean Diameter =", round(mean, 4), "mm")
print("Mean Error =", round(mean_error, 4), "mm")
print("Median =", round(median, 4), "mm")
print("Variance =", round(variance, 6), "mm2")
print("Standard Deviation =", round(std_dev, 6), "mm")
print("Coefficient of Variation =", round(cv, 4), "%")

print("\n===== QUARTILES =====")
print("Q1 =", q1)
print("Q2 =", q2)
print("Q3 =", q3)
print("IQR =", q3 - q1)

print("\n===== OUTLIER ANALYSIS =====")
print("Lower Bound =", lower_bound)
print("Upper Bound =", upper_bound)
print("Outliers =", outliers)
```

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```

# Histogram
plt.figure(figsize=(8,5))
plt.hist(data, bins=8)
plt.title("Histogram of Shaft Diameter")
plt.xlabel("Diameter (mm)")
plt.ylabel("Frequency")
plt.show()

# Box Plot
plt.figure(figsize=(6,4))
plt.boxplot(data)
plt.title("Box Plot of Shaft Diameter")
plt.ylabel("Diameter (mm)")
plt.show()
    
```

iv) Results and analysis from the graph

```

Mean Diameter = 50.0087 mm
Mean Error = 0.0087 mm
Median = 50.01 mm
Variance = 0.000469 mm2
Standard Deviation = 0.021663 mm
Coefficient of Variation = 0.0433 %
    
```

===== QUARTILES =====

```

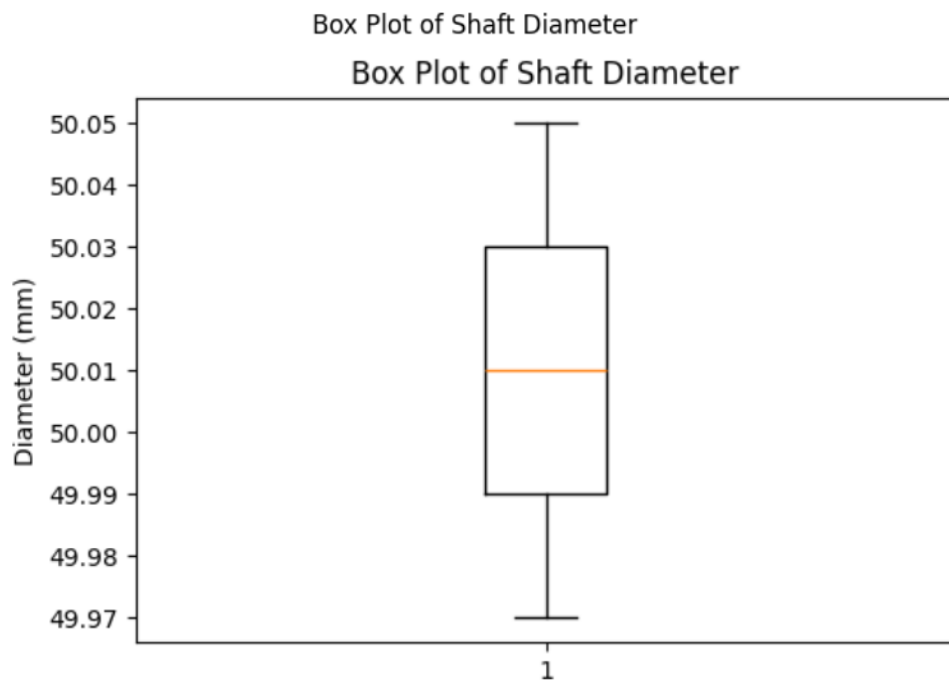
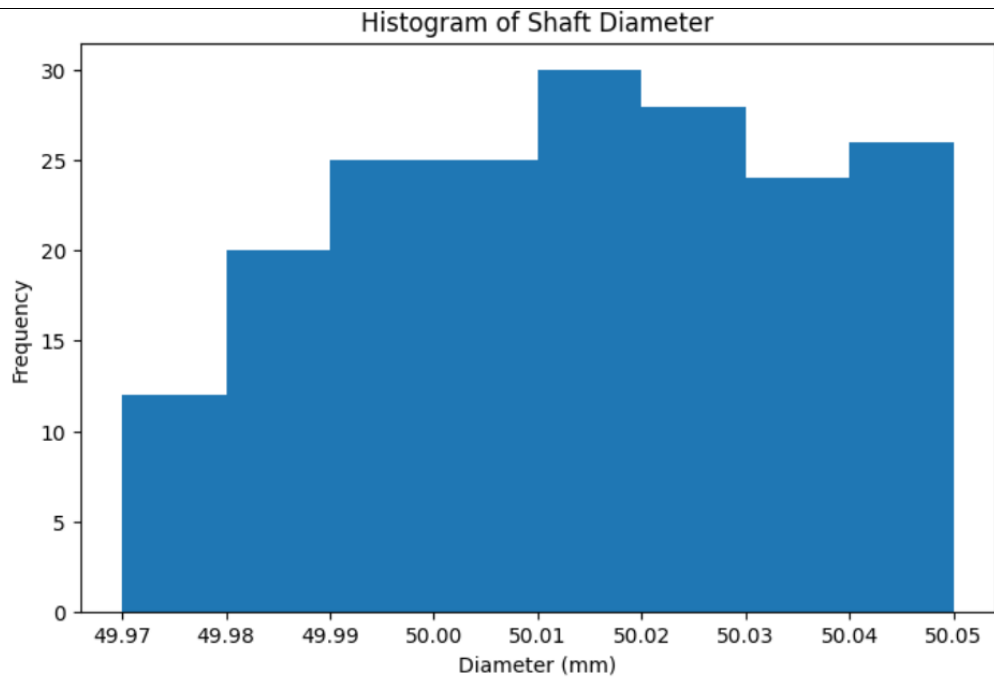
Q1 = 49.99
Q2 = 50.01
Q3 = 50.03
IQR = 0.039999999999999915
    
```

===== OUTLIER ANALYSIS =====

```

Lower Bound = 49.930000000000001
Upper Bound = 50.09
Outliers = []
    
```

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- The CNC machining process is highly accurate because the mean diameter (50.0087 mm) is very close to the target diameter (50.00 mm), resulting in a very small mean error (0.0087 mm).
- The process exhibits excellent precision and repeatability, as indicated by the low standard deviation (0.02166 mm) and very low coefficient of variation (0.0433%).

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	• The process appears statistically stable because the measurements are consistently distributed around the target value and no statistical outliers were detected using the IQR method. • The machine is capable of maintaining the specified tolerance (50.00 mm $\pm$ 0.05 mm) since all measured diameters fall within the acceptable range of 49.95 mm to 50.05 mm.